

# COMPENG 4TL4 – Lab 1 Report

## Introduction to Digital Signals Using MATLAB

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**Section:** L03

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## Point Sampling of a Sinusoid

- **Method:** A sinusoid of the form

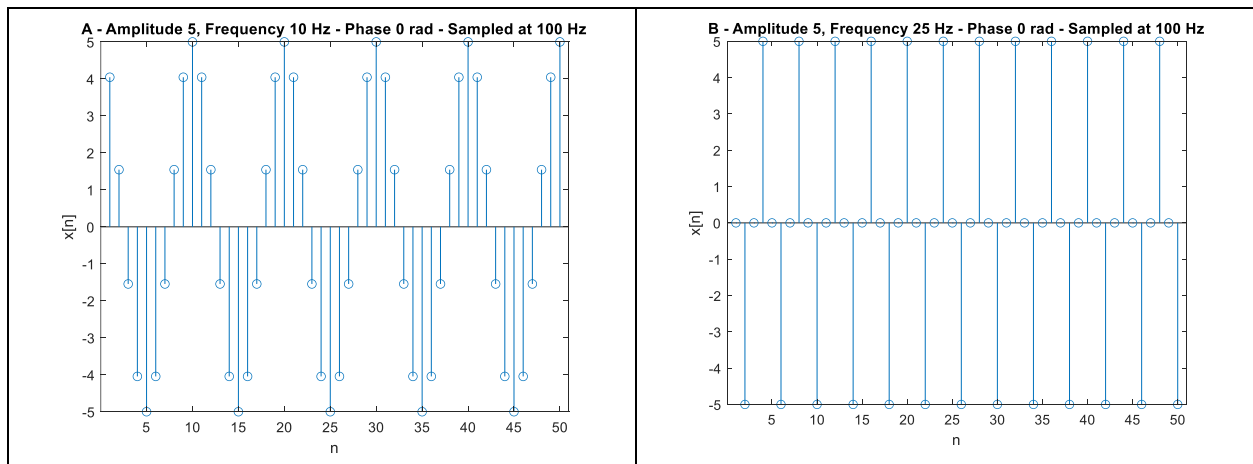
$$x(t) = A \cos(2\pi ft + \varphi)$$

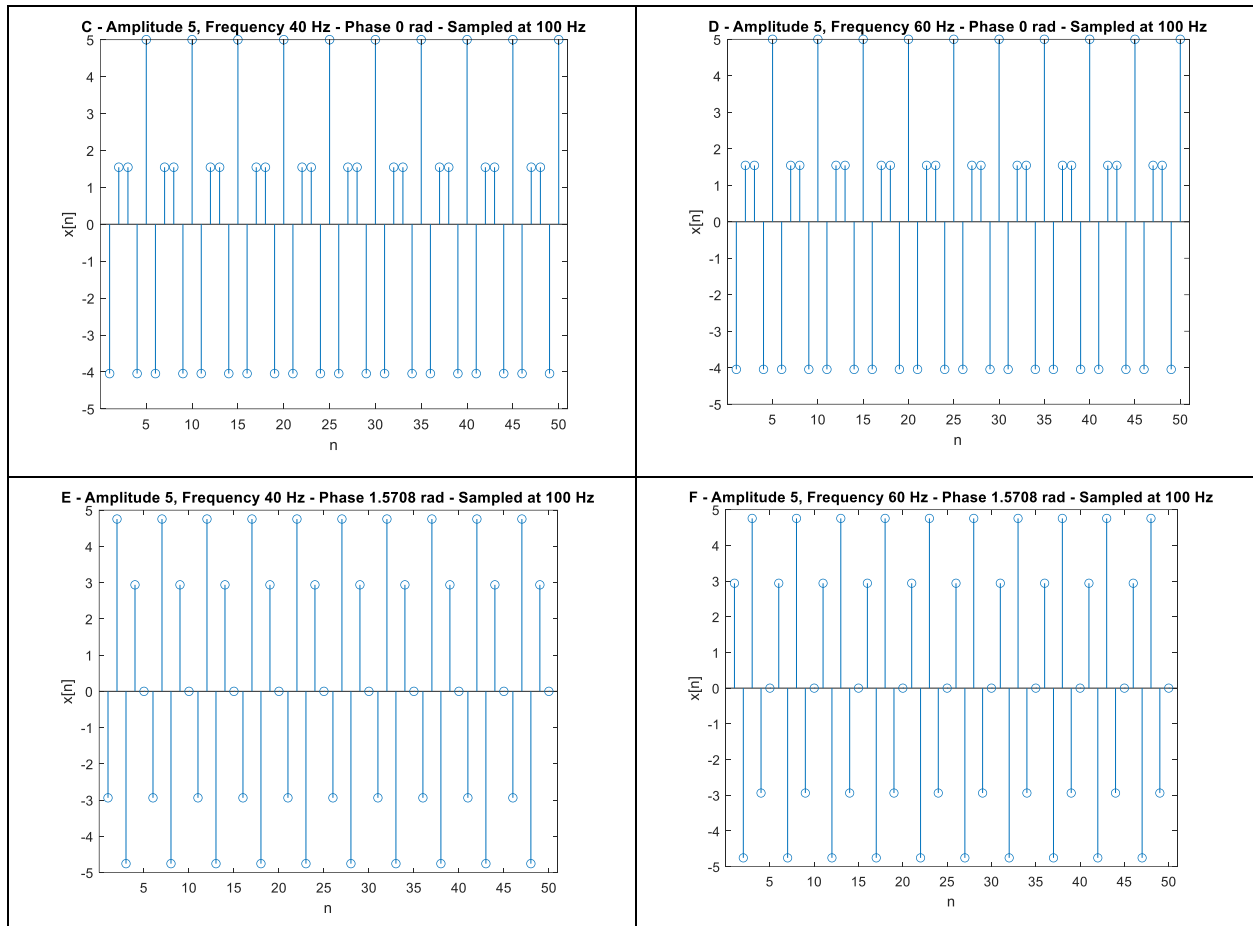
was sampled at 100 Hz for  $t = 0$  s to 0.5 s.

- **Cases Tested:**

- A:  $A=5$ ,  $f=10$  Hz,  $\phi=0$
- B:  $A=5$ ,  $f=25$  Hz,  $\phi=0$
- C:  $A=5$ ,  $f=40$  Hz,  $\phi=0$
- D:  $A=5$ ,  $f=60$  Hz,  $\phi=0$
- E:  $A=5$ ,  $f=40$  Hz,  $\phi=\pi/2$
- F:  $A=5$ ,  $f=60$  Hz,  $\phi=\pi/2$

- **Plots:**





The above plots are the sampled versions of signal A to F, where the signals' frequencies and phases were varied.

- **Observations:**

- A: Period after sampling =  $10 * T_s = 0.1s$
- B: Period after sampling =  $4 * T_s = 0.04s$
- C: Period after sampling =  $5 * T_s = 0.05s$
- D: Period after sampling =  $5 * T_s = 0.05s$
- E: Period after sampling =  $5 * T_s = 0.05s$
- F: Period after sampling =  $5 * T_s = 0.05s$

- **Conclusion:**

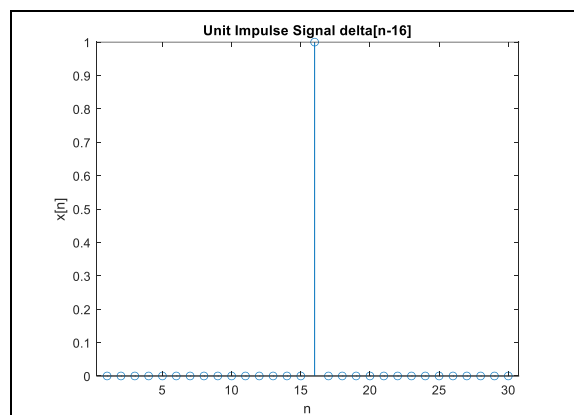
For signals A and B, the period of their discrete-time signals are 0.1s and 0.04s, which are the same as the period of the continuous time signal ( $1/10=0.1s$  and  $1/25=0.04s$ ). Because these signals have frequencies less than half of the sampling frequency (Nyquist sampling theorem) and the sampling frequency is also a multiple of their continuous time frequencies, it allows the signal's real

frequency/period to be represented even with discrete-time. For signal C and E where the period of the continuous time signals is  $1/40=0.025\text{s}$ , the period of the discrete-time signals is  $0.05\text{s}$ . This discrepancy is not due the sampling frequency being too low since  $100\text{ Hz}$  is still greater than  $2*40\text{ Hz}$ . Rather, this is because  $100$  isn't a multiple of  $40$ , so the discrete-time samples cannot align perfectly with the continuous-time cycles. As a result, the sampled signal exhibits an *apparent* period that is longer than the true continuous-time period. In other words, the discrete-time representation wraps around to the nearest integer number of samples, creating a periodicity of  $0.05\text{s}$  instead of  $0.025\text{s}$ . This effect is not aliasing but a consequence of the sampling frequency and the signal frequency not being harmonically related. If the plots for signal C and E are examined closely, it could be seen that the signal starting positive, goes negative, then positive, then negative again and then positive again within one period, effectively showing that two full cycles of the original continuous-time waveform are required before the discrete-time samples repeat. For signals D and F, which have a period in continuous-time of  $1/60=0.0167\text{s}$ , their period in discrete-time is  $0.05\text{s}$ . This is the same as signals C and E, despite signals D and F having a frequency of  $60\text{ Hz}$  instead of  $40\text{ Hz}$ . When examining the plot for signals D and F, it can be noted that the waveform looks the same (or similar in the case where phase is not equal zero) as that of signals C and E. This is due to aliasing and the Nyquist sampling theorem confirms this because  $100\text{ Hz}$  is less than  $2*60\text{ Hz}$ .

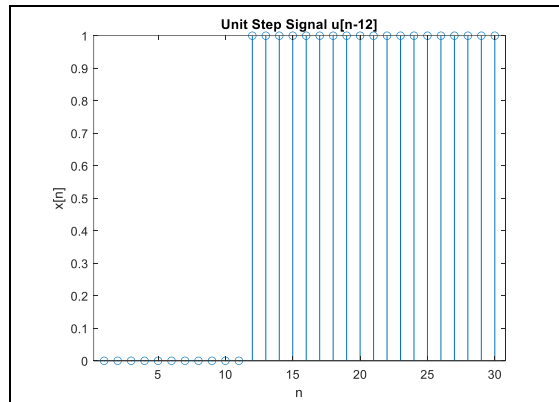
The phase change between signals C and E, and signals D and F, while keeping all else constant, results in a completely different discrete-time plot. This is because when there is a time shift, each sample may not necessarily be at the same part of the signal as with no time shift. This difference in alignment of samples and the continuous-time signal causes the discrete-time signal to look different. If a continuous cosine wave were to be fit on to all points of discrete-time signals C and E, it should be true that the cosine waves for signals C and E will have the same amplitude and frequency. The same applies for signals D and F.

## Unit Impulses and Steps

- **Impulse:**  $\text{delta}[n-16]$  was generated, showing a single spike at  $n=16$ .

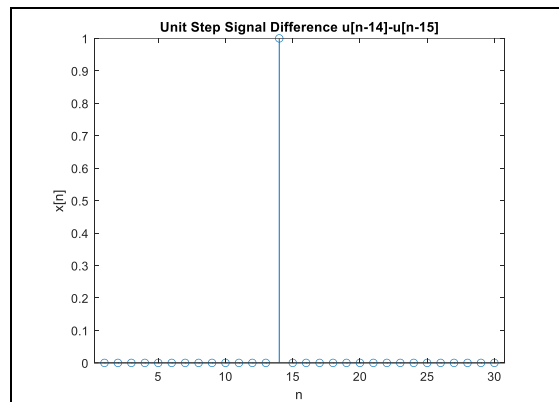


- **Step:**  $u[n-12]$  was generated, showing a transition at  $n=12$ .



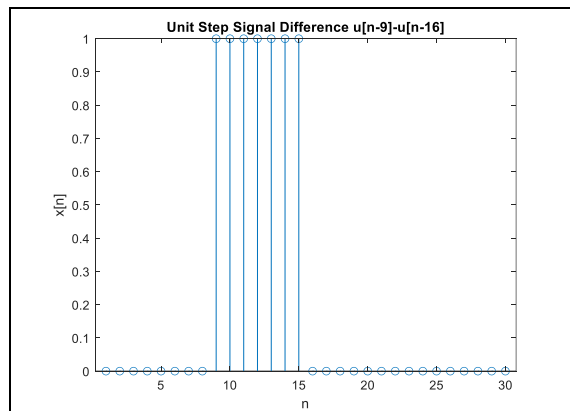
- **Difference Signals:**

- $x_1[n] = u[n-14] - u[n-15] = \text{delta}[n-14]$ .



This signal can be interpreted as  $\text{delta}[n-k]$  where  $k=14$ , since a spike with amplitude 1 can be observed at  $n=14$ .

- $x_2[n] = u[n-9] - u[n-16]$ , producing a rectangular pulse of width 7 samples.



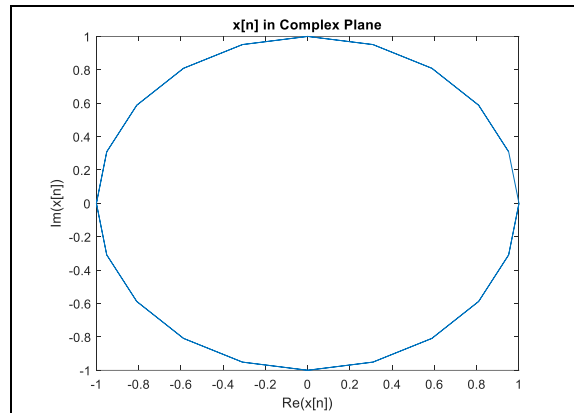
The resulting rectangular “pulse” has a width of  $15-9+1=7$ . This signal can be expressed as a sum of 7 unit impulses:  $x_2[n] = \text{delta}[n-9] + \text{delta}[n-10] + \text{delta}[n-11] + \text{delta}[n-12] + \text{delta}[n-13] + \text{delta}[n-14] + \text{delta}[n-15]$ .

# Complex Signals

- **Signal:**  $x[n] = Ae^{j\omega n}$ ,  $\omega = \frac{\pi}{10}$ ,  $A = 1$ ,  $n = 1$  to 40.

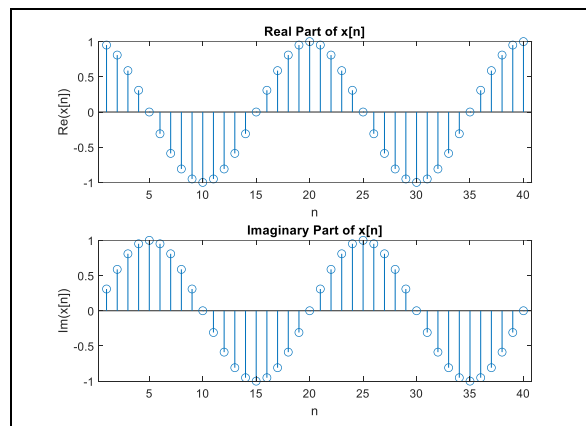
- **Plots:**

- Complex plane:



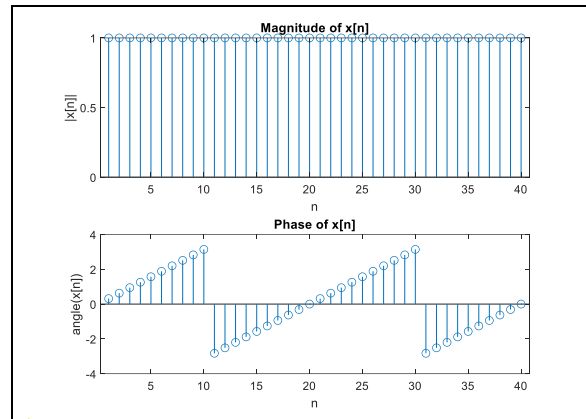
The plotted signal in the complex plane looks like a circle.

- Real and imaginary parts:



The real signal is a cosine wave with  $f=1/20=0.05$  and  $\omega=2*\pi*f=\pi/10$ , and the imaginary signal is a sine wave with  $f=0.05$  and  $\omega=\pi/10$ .

- Magnitude and phase:



Plot of the magnitude and phase of the signal where the magnitude is a constant 1 and the phase is a linear line which repeats every 20 samples.

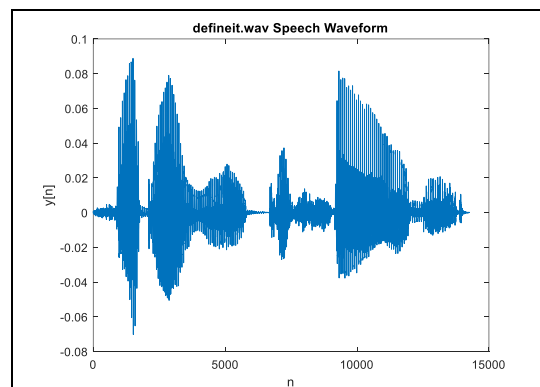
- **Conclusion:**

The plot of the magnitude makes sense because the signal  $Ae^{j\omega n}$  always has magnitude A (since  $|e^{j\omega n}| = 1$  regardless of the value of omega or n). A different value of n changes the phase of  $e^{j\omega n}$  but not the magnitude. The slope of the phase versus sample number curve is  $\pi/10$  and this is equal to the parameter omega. This means the slope of the phase versus sample curve is omega.

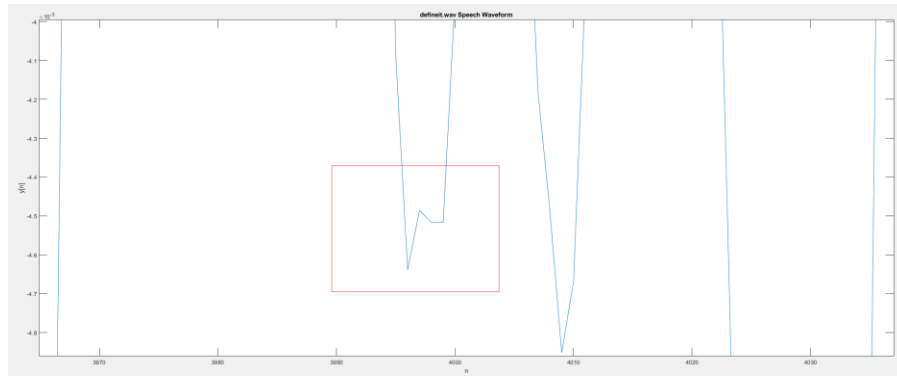
## Quantization of a Speech Signal

- **Original Signal:**

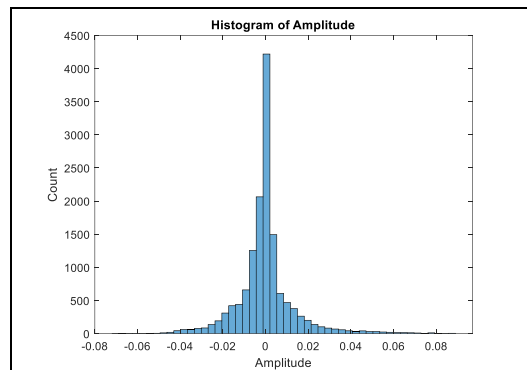
- Speech waveform:



The waveform shows a signal with many oscillations and varying amplitudes. When looked closely, the frequencies can be seen varying throughout the signal. The .wav file was quantized with 16 bits. Because this is still a relatively high precision, the amplitudes in the speech waveform still have many distinct levels it can take and does not visually look like a quantized signal. However, when zoomed in on the speech waveform, the distinct levels can be seen.



- Histogram of amplitude:

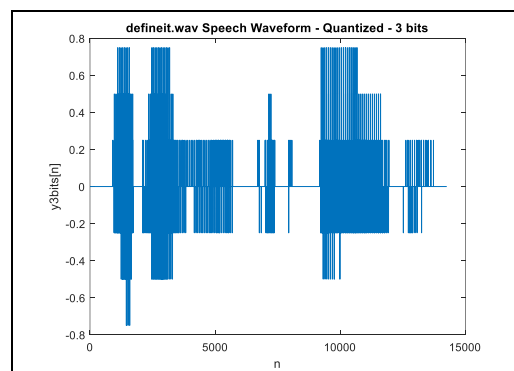


The shape of the histogram resembles a Gaussian distribution with mean zero, since the majority of the samples have amplitude near zero.

The sound quality of the speech signal is clear and clean from noise. The words being spoken can easily be heard and understood.

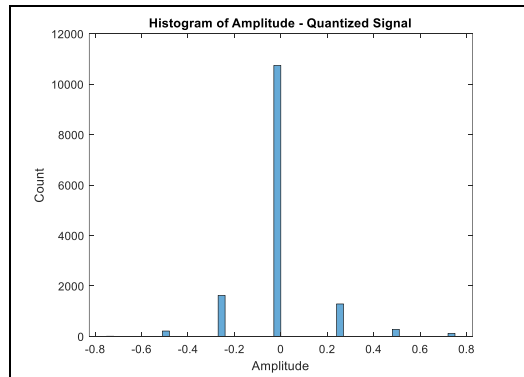
- **3-bit Quantization:**

- Speech waveform:



The waveform shows a signal that maintains the overall shape of the original signal, however, it is evident that a lot of information has been lost. Due to quantization, the distinct levels that the amplitude can take can be seen.

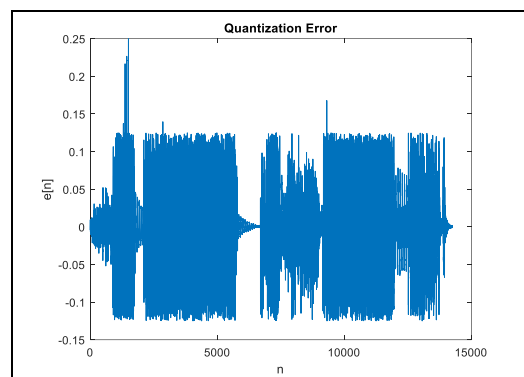
- Histogram of amplitude:



The shape of the histogram still resembles a Gaussian distribution with mean zero. Since the majority of the samples with amplitude near zero and the quantizer has a zero bin, the histogram has a high count at zero. In addition, the number of available amplitudes values in the histogram has decreased due to quantization.

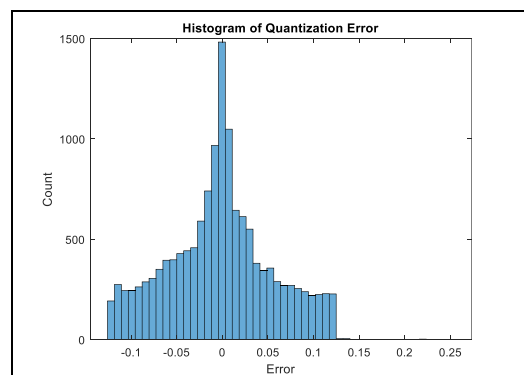
The sound quality has degraded heavily, and noisy and staticky sounds can be heard. However, the words being spoken remains intelligible.

- Waveform of the error:



The majority of quantization error value can be seen between 0.125 and -0.125, with the highest error reaching 0.25.

- Histogram of the error:

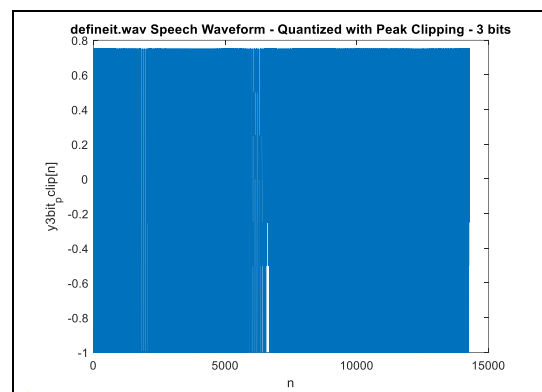


The distribution of the quantization error can be seen more clearly in the above histogram. The quantization error being between 0.125 and -0.125 makes sense because with 3-bit quantization, there are  $2^3=8$  bins, with bin size equal  $\text{range}/\# \text{ of bins} = 2/8=0.25$  (range is from -1 to 1), and with the levels

placed in the middle of the bin, the majority of the error magnitudes should be less than  $0.25/2=0.125$ . It can be noted that the maximum quantization error is actually 0.25. This is because the quantizer shifts the bins by 0.125 to the left to create a bin with zero as the level, leading to one bin with level 0.75, which will have a quantization error of 0.25 if the signal has amplitude 1. Because the distribution of the signal is Gaussian centered around zero and the quantizer has a zero bin, the quantization error will be equal to the amplitude of the signal for samples with amplitude near zero. And because the amplitude of the samples is near zero, the quantization error will also be near zero. This leads to a distribution of the quantization error heavily weighed at zero. Because of the distribution of our signal, this quantizer has a lower quantization error compared to the standard quantizer without the bin shift, on average. However, the maximum error is higher.

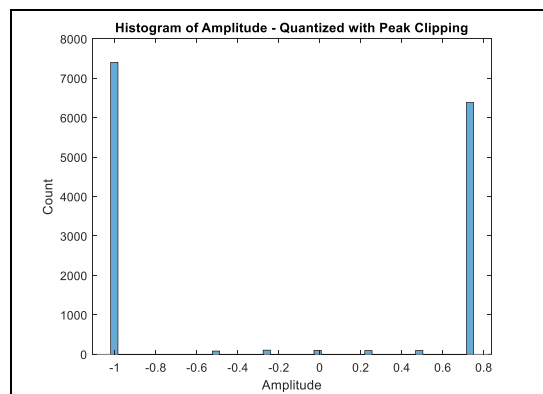
- **3-bit Quantization with Heavy Peak Clipping:**

- Speech waveform:



The waveform shows a signal that is completely different from the original signal. The amplitude seems to be almost purely oscillating between the highest level (0.75) and the lowest level (-1).

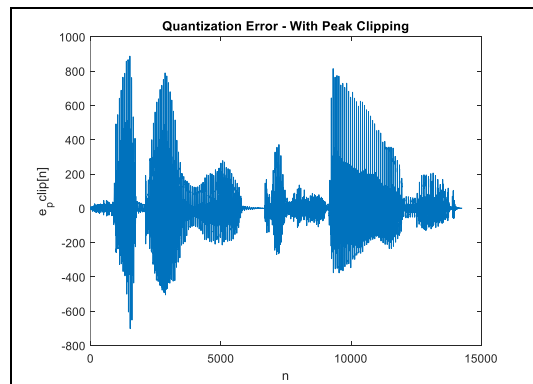
- Histogram of amplitude:



The shape of the histogram confirms this as two bins at -1 and 0.75 can be seen taking the majority of the amplitude values with very little amplitudes falling in between. This makes sense because if the majority of the signal samples are scaled to an amplitude with magnitude greater than 1 (outside the quantizer range of -1 to 1), peak clipping occurs and those majority of samples simply take on either the highest level or the lowest level.

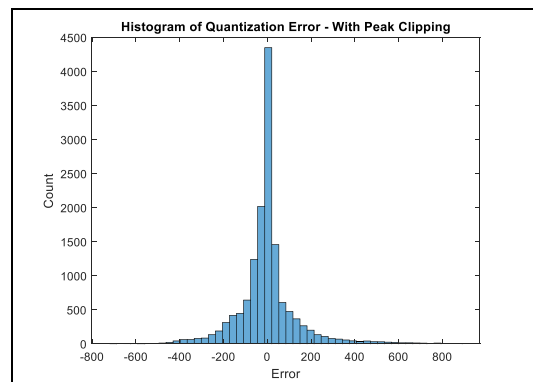
The sound quality has degraded even further, with lots of noisy and staticky sounds. Surprisingly, the words being spoken can still be heard and somewhat still remains intelligible.

- Waveform of the error:



The waveform of the quantization error looks almost identical to the waveform of the original speech waveform. This makes sense because the quantization error equals the original amplitude minus the quantized amplitude, and if the quantized amplitude is just a constant 0.75 for positive values and -1 for negative values, the error is simply just the original amplitude shifted down/up. Because the amplitude is scaled by 10,000, an amplitude shift of 0.75 or 1 becomes negligible. The difference between this error waveform and the original speech waveform is the amplitude values.

- Histogram of the error:



The histogram of the error also looks exactly like the histogram of the original speech signal. This makes sense because their waveforms are identical.